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Updated Database of Clothing Thermal Insulation and Vapor Permeability Values of Western Ensembles for Use in ASHRAE Standard 55, ISO 7730, and ISO 9920

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This paper is based on findings resulting from ASHRAE Research Project RP-1760.

ABSTRACT

ASHRAE Research Project RP-1760 aimed to update the database of western clothing as used in ANSI/ASHRAE Standard 55-2013, Thermal Environmental Conditions for Human Occupancy *(ASHRAE 2013a), ISO Standard 7730-2005,* Ergonomics of the Thermal Environment—Analytical Determination and Interpretation of Thermal Comfort Using Calculation of the PMV and PPD Indices and Local Thermal Comfort Criteria *(ISO 2005), and ISO Standard 9920-2009,* Ergonomics of the Thermal Environment—Estimation of Thermal Insulation and Water Vapor Resistance of a Clothing Ensemble *(ISO 2009). The previous database, established in the 1970s and 1980s, relied mostly on single-zone manikins and did not provide detail on air and body movement effects on insulation. Insulation values of up to 70 clothing ensembles (31 male, 39 female) were measured in a static standing posture at 0.2, 0.4, and 1 m/s^{-1} (0.66, 1.31, and 3.28 ft/s^{-1}) air speed, walking in 0.2 and 1.0 m.s^{-1} air speed, and in a sitting posture in 0.2 m.s^{-1} air speed. Measurements were conducted in three laboratories on three manikins with up to 34 individually controlled zones. In addition, vapor resistance was determined in 31 ensembles. This new database provides total, intrinsic and air insulation values, vapor resistance, and clothing area factors as well as correction factors to estimate the wind and movement effect on clothing insulation. Furthermore, the model to estimate the clothing area factor from insulation values of the ensembles was updated reflecting changes in commonly worn clothing styles.*

INTRODUCTION

Clothing insulation is a crucial parameter in the assessment of thermal comfort (Fanger 1972; Havenith et al. 2002; Al-Rashidi et al. 2009, 2010; ASHRAE 2009) and, therefore, important for the optimal design of the built environment (Ahmad and Ibrahim 2003; Cheong et al. 2003; Kwok and Chun 2003). Though a substantial data set on western clothing already exists (Olesen and Nielsen 1983; McCullough and Jones 1984; McCullough et al. 1985), these data originate mainly from the 1980s. At the time, spatial resolution of thermal manikins was limited with many designed as a single zone (some with separate control of extremities) with a limited number of spot skin temperature sensors used to determine mean skin temperature (McCullough and Jones 1984). Furthermore, no spatial (i.e., regional zone) data on clothing insulation were reported, limiting the utility of the data for modeling purposes.

Given the need to curb growth in energy demands, precise knowledge of the clothing insulation parameter is essential in optimizing the design of ventilation and air-conditioning systems. Predictions of human thermal comfort sensations (e.g., predicted mean vote [PMV]/percentage dissatisfied [PPD]) require information on the value of the clothing insulation as worn by occupants. Whilst the adaptive approach to thermal comfort (currently optional in ASHRAE Standard 55 [2013a]) might not require values for clothing thermal insulation to assess whether an environment might be acceptable, this approach applies only to environments that are naturally conditioned, without mechanical cooling or heating in operation, and when the outdoor temperature range is 10°C to

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33.5ºC (50°F to 92ºF). Beyond these conditions, alternative approaches will still be required, thus rendering data relating to clothing thermal insulation values essential. Furthermore, such data are relevant beyond comfort per se, and also inform both heat and cold stress standards (ISO 2004a, 2007).

A vast amount of spatial information relating to insulation distribution can now be gained from modern techniques, such as the use of new generation thermal manikins. High-resolution data have recently been published for non-Western ensembles (Havenith et al. 2013, 2015) and provide detail on static whole-body clothing insulation; effects of body posture and movement (e.g., sitting, standing, walking); effects of air movement on clothing thermal insulation; local insulation for different body parts including changes due to movement, posture, and air movement; clothing vapor permeability; and clothing surface area factors. These detailed data currently relate to non-Western ensembles only and were originally aimed at users of ASHRAE Standard 55 (2013a) and ISO 7730 (2005). These data have now also been adopted for work with more complex models (e.g., computational fluid dynamics [CFD]) that require more detailed information on the boundary conditions for different body parts.

In contrast, the data currently available on Western clothing are far less comprehensive. Also, new developments in modern Western apparel relating to both design style and the inclusion of new technical fibers (e.g., fleeces made of polyethylene, polyamide, or polyester) must be considered. To bridge the gap between existing data relating to non-Western and modern Western clothing, ASHRAE commissioned a research project, titled "Update of Clothing Database for Existing and New Western Clothing Ensembles, Including Effects of Posture, Body and Air Movement" under code 1760-TRP. The work was commissioned to a consortium of three research laboratories: the Environmental Ergonomics Research Centre (EERC) at Loughborough University, Loughborough, U.K.; the Department of Design Sciences, Ergonomics and Aerosol Technology (EAT), at Lund University, Lund, Sweden; and the Central Institute for Labour Protection—National Research Institute (CIOP-PIB) in Warszawa, Poland. The information gathered through this research will constitute an important new addition to the data in building comfort standards. This additional information will enable users of ASHRAE Standard 55-2013 (ASHRAE 2013a), *ASHRAE Handbook—Fundamentals* (ASHRAE 2013b), ISO Standard 7730-2005 (ISO 2005), ISO Standard 9920-2009 (ISO 2009), and those using alternative approaches (e.g., CFD modeling) to make more accurate thermal comfort predictions under realistic conditions for people clothed in Western attire as applicable for buildings, aircraft, rail, and road vehicles.

Data pertaining to the static thermal insulation values for the standing and sitting postures will be reported in this paper. In addition, vapor resistance values for the standing posture will be provided. Given the large amount of data to report, local insulation data and data collected on the effects of body movement and air movement on the thermal insulation values of non-Western clothing will be reported in a subsequent paper.

METHODS

Surveys were conducted of commonly worn clothing in the U.S., Japan, Australia, and the U.K. (Smallcombe et al. 2020). From these, a total of 71 ensemble configurations were selected. A detailed description (including photos) of the tested clothing ensembles is provided in the appendix. The experimental work was performed using thermal manikins in the three separate laboratories.

The thermal manikins used at these research institutions are presented in Figure 1. The procedures described in ISO 15831 (2004b) and ISO 9920 (2009) were followed for all manikins unless specifically stated otherwise. By controlling the skin temperature of the manikin regions at 34ºC (93.2ºF) and fixing the environmental conditions, the measured heat loss rate can be used to calculate dry heat resistance of the clothing worn. Each ensemble was measured at least twice to ensure accuracy, with the mean values taken. If the discrepancy between measurements was >4%, a third measurement was taken (ISO 2004b). Total insulation values were calculated according to the parallel method (ISO 2004b, Havenith 2005).

Environmental Ergonomics Research Centre, School of Design and Creative Arts, Loughborough University, U.K.

At the EERC, all 71 ensembles selected were measured for dry heat resistance in standing and sitting postures. Using the "Newton" manikin (developed by Thermetrics, Seattle, USA), which is constructed of thermally conductive, copper-filled carbon-epoxy with embedded heating wire elements (with a maximum output 800 W/m^2) and distributed wire sensors in 34 microprocessor-controlled heating zones (Figure 1a).

For the sitting conditions, a mesh chair was used to show the effect of the posture and the clothing compression by the chair without adding potential insulation or vapor resistance from a chair.

Dry heat resistance of the ensembles was determined in either 21ºC (69.8ºF) or 18ºC (64.4ºF), for higher insulating clothing, at 50% relative humidity. (Any variations were recorded, and calculations of results accounted for these). Mean radiant temperature was equal to air temperature (confirmed beforehand by black globe measurement). For each ensemble, insulation was measured in both standing and sitting postures (Figure 1a) at an air velocity of 0.2 m/s^{-1} (0.66 ft/s^{-1}).

All measurements were conducted in a custom-built wind tunnel within a climate-controlled room (Havenith et al. 2015). The air velocities were accurately determined in a series of pilot tests prior to the main experiments. The actual measured average air velocity at the location of the manikin (tested at three heights) was 0.21±0.01 m/s^{-1} (0.69 ft/s^{-1}). During manikin testing, air temperature, globe temperature, relative humidity, and

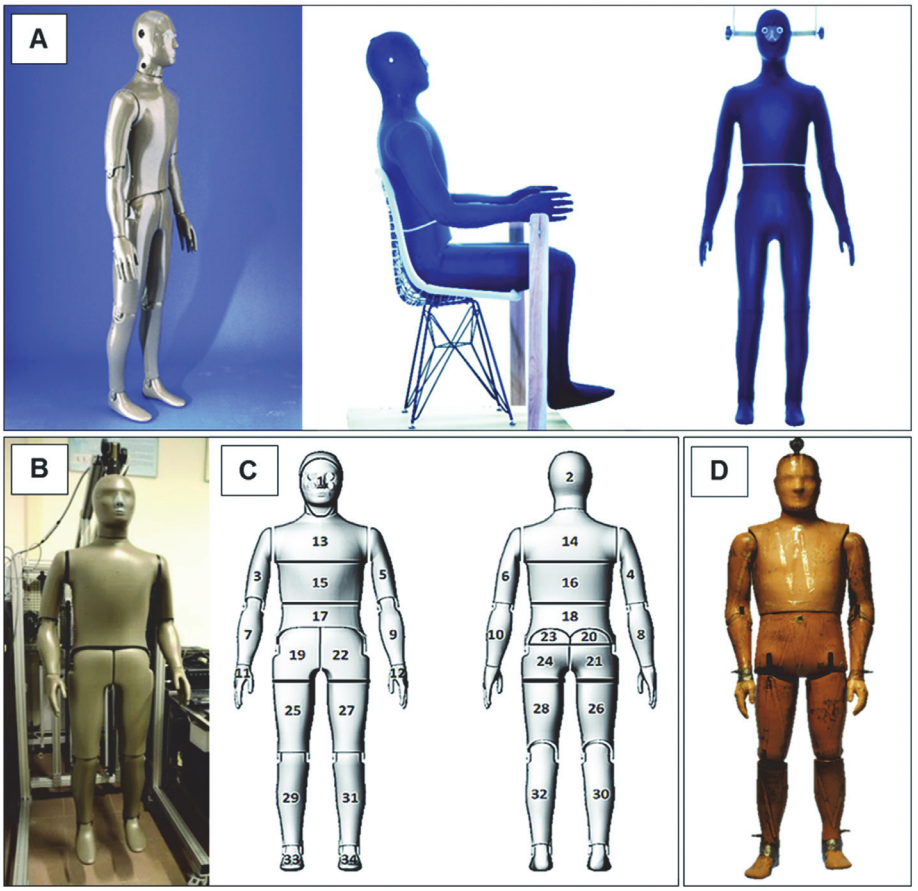


Figure 1 *a) The "Newton" manikin used at Loughborough University in standing and sitting postures, b) the "Newton" manikin used at CIOP-PIB Poland, c) depiction of regional zones for both Newton manikins, and d) The "Tore" manikin used at EAT Lund University.*

air velocity were measured using a Testo 454/350 unit (Testo AG, Testostraße 1, 79853 Lenzkirch, Germany) with unidirectional vane and hot wire sensors placed at the correct angle to the airflow direction. The mean radiant temperature was assumed to be equal to the air temperature based on comparisons of air temperature and black globe temperature/mean radiant temperature (Testo) in the chamber. The manikin system includes two PT100 sensors (Betatherm 30K5A1B, Ireland) for air temperature at a 0.3 m and 1.4 m (1 ft and 4.6 ft) height and a Vaisala Humidity sensor (HUMITTER 50U).

The manikin was calibrated according to the procedure described in ASTM Standard F1291-10 (ASTM 2010). Manikin skin zone sensors were calibrated with the manikin heating deactivated, placed horizontally in a climatic chamber in a turbulent high air velocity (>4 m/s^{-1}[13.1 ft/s^{-1})] using the certified Testo high-precision sensors as reference. To prevent air penetrations of manikin joints, all dry measurements were conducted with the manikin fitted with a tight-fitting textile (polyester) "skin." The regional dry insulation of this skin (whole body average 0.02 m^2·K·W^{-1} (0.13 clo) was established at the start of testing and reverified at the end of the project, and was subtracted from the regional dry insulation values obtained for all clothing ensembles. The manikin temperatures and heat losses were integrated over and recorded at 30 second intervals. A minimum of 30 minutes of measurement stability was required. Of these, data from the last 15 minutes of the steady-state period were used for the insulation calculation.

Ergonomics and Aerosol Technology, Department of Design Sciences, Lund University, Lund, Sweden

At this research laboratory, a subselection of 35 clothing ensembles were tested for dry insulation in the standing and sitting postures. The male-shaped thermal manikin "Tore" is made of plastic with a metal frame inside to support the body parts and joints. Tore is divided into 17 individually controlled zones: head, chest, back, stomach, buttocks, left and right upper arm, left and right lower arm, left and right hand, left and right thigh, left and right lower leg, and left and right foot (Figure 1d). For the sitting conditions, a mesh chair was used, to show the effect of the posture and the clothing compression

by the chair, but without adding potential insulation or vapor resistance from a chair.

Air velocity was measured with Swema Air 3000d with omnidirectional SWA 03 sensor (Swema AB, Sweden). Ambient air temperature was continuously monitored using three sensors (PT100, Pico Technology Ltd., U.K.) positioned adjacent to the ankles, the mid-trunk, and the head (vertical heights of 0.1, 1.1, and 1.7 m [0.3, 3.6 and 5.6 ft], measured from the soles of the manikin, respectively).

The air temperature in the chamber was set to 22.2°C±0.1ºC (72.0°F±0.2ºF). Due to the characteristics of the chamber used, the mean radiant temperature was assumed to be equal to the air temperature. The air velocity in the chamber was 0.21 m/s^{-1}±0.07 m/s^{-1}(0.69 ft/s^{-1}±0.23 ft/s^{-1}). The manikin temperatures and heat losses were recorded at 10 second intervals. A minimum of 20 minutes of measurement stability was required. Of these, data from the last 10 minutes of the steady-state period were used for the insulation calculation. Calibration and validation trials at Lund University confirmed that this 10-minute steady-state period provided repeatable and reliable results. Offset calibration of the manikin's surface temperature sensors was carried out in the homogenous conditions (34°C [93.2°F]) in a warm chamber with the same Pico equipment as stated previously. Standard reference clothes of the laboratory and nude manikin were measured before the measurement series to confirm the same results as measured previously.

Central Institute for Labour Protection—National Research Institute (CIOP-PIB), Warszawa, Poland

At this laboratory, the dry insulation of a subselection of 31 clothing ensembles was determined in the standing posture using a "Newton" thermal manikin consisting of 34 microprocessor-controlled heating zones (Figure 1b). Dry heat resistance of the ensembles was determined in either a 20ºC (68.0ºF) or a 15ºC (59.0ºF), for higher insulating clothing, environment at 40% relative humidity. (Any variations were recorded, and calculations of results accounted for these). To prevent air penetration of manikin joints, all dry measurements were conducted with a tight-fitting textile "skin" of polyamide (80%) and elastin (20%). The dry insulation of this skin (0.02 $m^2 \cdot K \cdot W^{-1}$) was established at the start of testing and reverified at the end of the project, and was subtracted from the dry insulation values obtained for all clothing ensembles.

In addition, the evaporative resistance of these 31 clothing ensembles was determined in a climatic chamber in isothermal conditions. All (wet) evaporative resistance measurements were conducted at a temperature of 34ºC (93.2ºF), 40% relative humidity with 0.2 m/s^{-1} (0.67 ft/s^{-1}) air speed, starting with a prewetted skin and with the manikin sweat rate set to 300 $mL \cdot h^{-1} \cdot m^{-2}$ (Młynarczyk 2019). In this isothermal environment, no or minimal dry heat loss is present, making the measurement of evaporative heat loss more accurate.

For the (wet) evaporative resistance measurements, the manikin was fitted with a tight-fitting, moisture-absorbing, textile "skin" of polyamide (80%) and elastin (20%). This skin was wetted with distilled water to simulate a sweating skin surface. The water evaporates from the manikin through the garments and draws heat of evaporation from the manikin. Given the skin temperature, the water vapor pressure at the skin and the vapor pressure gradient between skin and environment can be calculated. In combination with the heat loss from the manikin, this allows derivation of the clothing's evaporative heat resistance. No correction for possible cooling of the textile skin was applied, consistent with the standard (ISO 2004b). For all evaporative resistance calculations, the parallel calculation method was used (see the "General Methods" section of this paper).

Throughout testing, the ambient environmental conditions were measured by a microclimate sensor (EHA MM10, Ekohigiena Aparatura, Poland) and a 1221 Thermal Comfort Data Logger (INNOVA AirTech Instruments, Denmark). The manikin system utilized a Vaisala humidity sensor (Finland) and PT100 sensors (Betatherm, Ireland) for ambient temperature. Manikin temperatures and heat losses were integrated over and recorded at 30 second intervals. A minimum of 30 minutes of measurement stability was required. Of these, data from the last 15 minutes of the steady-state period were used for the insulation calculation.

GENERAL METHODS

Clothing Surface Area Factor Measurement, f_{cl}

Clothing surface area measurements were based on the photographic method using a frontal and a side picture (Havenith et al. 2015). A projection of the area factor was used and analyzed in Adobe Photoshop. In general, the procedure follows advice from the literature (Seppanen et al. 1972; Sprague and Munson 1974; McCullough et al. 2005). Previously, several projects have assessed the area factors determined using different numbers of photographs and compared this technique with other methods (e.g. digital body scanning). The latter method is laborious and often requires considerable post scanning editing of images (folds in the clothing often require manual corrections to the datasets). McCullough et al. (2005) reported that for protective clothing, taking a single photograph from the front provided the same information as the more elaborate measurement via 6 pictures from different angles (r2=0.99, mean square error=0.015). As per the methods employed when assembling a clothing insulation database for non-western clothing (Havenith et al., 2015), the clothing surface area factor was quantified using a frontal and side-on photograph.

Air insulation, I_a

In all laboratories, the insulation of the nude manikin was determined for the value of the surface air insulation heat resistance, I_a.

Calculations for I_T and I_{cl}

For the dry measurements, the dry heat insulation I_T was determined using the parallel calculation method as (Havenith et al. 2002, 2005; Kuklane et al. 2007, 2012):

$$I_T = \text{dry heat resistance} = \frac{(T_S - T_o)}{\Sigma_{i=1}^{n}\left(\frac{A_i \times H_{c,i}}{A}\right)}[m^2 \cdot K \cdot W^{-1}] \quad (1)$$

where

T_s = mean temperature at the manikin surface, °C

T_o = operative temperature, °C

A = total area of the manikin's surface, m^2

$H_{c,i}$ = total power required for the manikin for the manikin segment, Wm^{-2}

A_i = surface area segment i (m^2)

The intrinsic clothing insulation I_{cl} was determined as:

$$I_{cl} = I_T - \frac{I_a}{f_{cl}}[m^2 \cdot K \cdot W^{-1}] \quad (2)$$

Conversion from SI units to clo units can be performed by:

$$I(\text{clo}) = \frac{I(\text{SI units})}{0.155} \quad (3)$$

For the wet manikin measurements (conducted at CIOP-PIB Poland), evaporative resistance was determined using the parallel method as defined as:

$$R_{e,T} = \frac{p_{sk} - p_a}{\Sigma_{i=1}^{n}\left(\frac{A_i \times H_{e,i}}{A}\right)} \quad (4)$$

where

$R_{e,T}$ = the total clothing evaporative resistance calculated by the parallel heat loss method, $kPa \cdot m^2 W^{-1}$

A, A_i = the total sweating surface area and segmental sweating surface area, respectively, m^2

i = the number of the segment of the sweating thermal manikin (i=1, 2,…, n);

p_{sk}, p_a = the water vapor pressure on the whole fabric skin surface and in the ambient air, respectively, kPa

$H_{e,i}$ = the evaporative heat loss of segment i, Wm^{-2}

The water vapor pressures at the fabric skin surface temperature (assumed to be fully saturated) and the air temperature were calculated by the Antoine's equation:

$$p_{sk} = exp\left(18.956 - \frac{4030.18}{t_{sk} + 235}\right) \times RH_{sk} \quad (5)$$

$$p_a = exp\left(18.956 - \frac{4030.18}{t_a + 235}\right) \times RH_a \quad (6)$$

where

t_{sk}, t_a = temperatures at the wet fabric skin surface and in the ambient air, respectively, °C

RH_{sk}, RH_a = the relative humidity at the wet fabric skin surface and in the ambient air, respectively, % (assumed that RH_{sk} on the saturated wet fabric skin surface was 100%)

p_{sk}, p_a = the water vapor pressure on the whole fabric skin surface and in the ambient air, respectively, kPa

Calculations for i_m

The observed values for $R_{e,T}$ and I_T were used to calculate the clothing vapor permeability index, i_m:

$$i_m = \frac{I_T}{L \cdot R_{e,T}} \quad (7)$$

where

L = Lewis constant, 16.5 K kPa^{-1}

To calculate $R_{e,T}$ from the reported i_m values the following equation can be used:

$$R_{e,T} = i_m \cdot L \cdot I_T \quad (8)$$

RESULTS AND DISCUSSION

Manikin Comparison

To determine the effect of using different thermal manikins in different climatic chambers, dry insulation data obtained from the laboratories in Poland and Sweden were compared to the insulation values for the same clothing ensemble obtained at Loughborough, U.K. (Figures 2 and 3).

The data presented in Figures 2 and 3 indicate good agreement between the insulation values obtained in the three different laboratories in a standing posture. On average, the values obtained in Sweden using the "Tore" manikin were slightly higher than in Loughborough (R^2 = 0.93, Δ = +5.3%, SD 4.4%). This variation can be related to the different manikin shape (Havenith et al., 2015) as this interacts with the clothing fit and drape, which subsequently affect the air layers in and surrounding the clothing, and secondly by differences in climatic chamber airflow patterns and thirdly by the use of a skin in Loughborough potentially affecting convective patterns. This will be discussed in more detail later. Comparing the data obtained in the Loughborough and Poland laboratories—which both used the same model of the "Newton" manikin, but in different chamber designs – the agreement between the dry insulation values was higher (R2 = 0.97; Δ =+0.7%, SD 3.0%), despite using different climatic chamber setups.

Similarly, despite the different manikins (Newton vs. Tore), good agreement (R^2 = 0.94, Δ = –0.8%, SD 3.6%) was observed between values obtained in EERC Loughborough

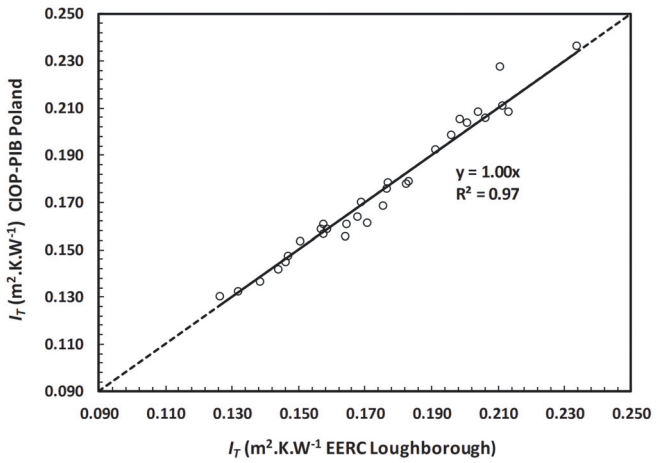


Figure 2 *Comparison of the dry insulation data measured on "Newton" manikin in CIOP-PIB Poland to the data obtained for the same clothing ensembles using the "Newton" in EERC Loughborough.Data presented are for the standing posture at air velocity of 0.2 m/s^{-1} (0.67 ft/s^{-1}).*

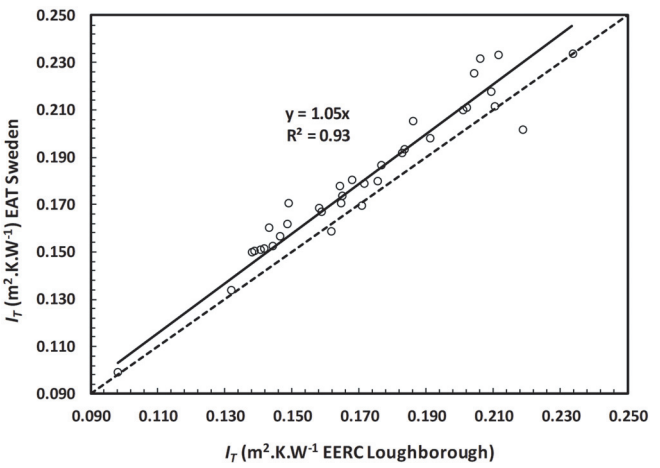


Figure 3 *Comparison of the dry insulation data measured on "Tore" manikin in EAT Sweden to the data obtained for the same clothing ensembles using the "Newton" in EERC Loughborough. Data presented are for the standing posture at air velocity of 0.2 m/s^{-1} (0.67 ft/s^{-1}).*

and EAT Sweden in the sitting posture at an air velocity of 0.2 m/s^{-1} (0.67 ft/s^{-1}) (see Figure 4).

To account for the difference in standing dry insulation values obtained in Loughborough ('Newton' manikin) and Sweden ('Tore' manikin), the standing values obtained in Loughborough were corrected by +2.65% to derive a value representative of the dry insulation quantified using the two different manikin designs. Due to negligible laboratory differences observed for the sitting posture, no correction was applied to these data and, thus, the sitting values reported are those obtained in Loughborough ('Newton' manikin).

Clothing Surface Area Factor

The f_{cl} values were used in the established way to calculate the intrinsic clothing insulation, I_{cl}. This model of intrinsic and surface air boundary layer insulation is based on a cylindrical model of clothing (Figure 5), which is typical for Western style clothing with all clothing worn as cylinders around torso and limbs.

In the various clothing publications dealing with the f_{cl} factor, prediction equations for f_{cl} are provided. These are based on Western clothing. One of the most commonly used equations is based on work by McCullough et al. (1985), which reads:

If I_{cl} is expressed in m^2·K·W^{-1}:

$$f_{cl} = 1.08 + 1.45I_{cl} \qquad R^2 = 0.55 \tag{9}$$

Or with a fixed constant:

$$f_{cl} = 1.00 + 1.97I_{cl} \qquad R^2 = 0.95 \tag{10}$$

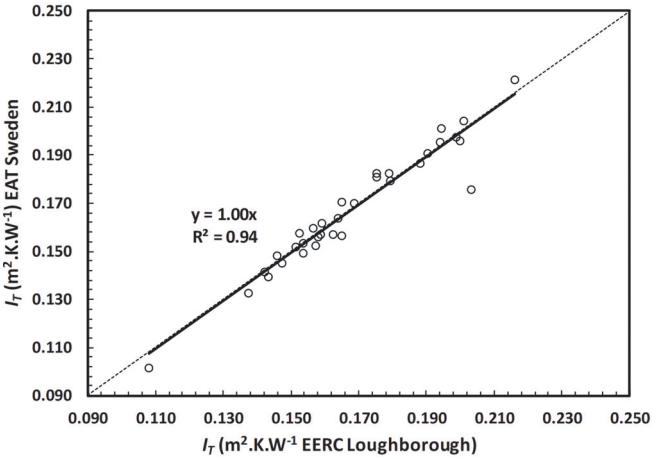


Figure 4 *Comparison of the dry insulation data measured on "Tore" manikin in EAT Sweden to the data obtained for the same clothing ensembles using the "Newton" in EERC Loughborough. Data presented are for the sitting posture at air velocity of 0.2 m/s^{-1} (0.67 ft/s^{-1}).*

If I_{cl} is expressed in clo:

$$f_{cl} = 1.08 + 0.225I_{cl} \qquad R^2 = 0.55 \tag{11}$$

Or with a fixed constant:

$$f_{cl} = 1.00 + 0.31I_{cl} \qquad R^2 = 0.95 \tag{12}$$

The application range for which these relations were tested is for I_{cl} between 0.2 and 1.7 clo.

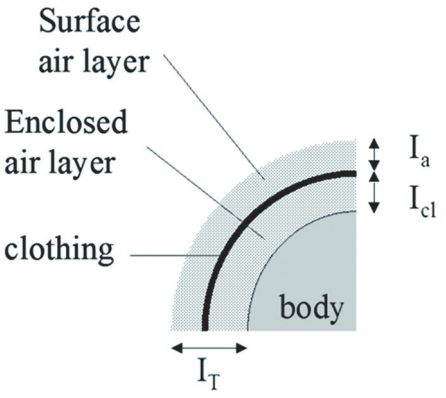


Figure 5 *Cylindrical model of total and intrinsic clothing insulation (ISO Standard 9920 [ISO 2009]).*

For the clothing studied in the present project (Figure 6), the equations (one with the constant fixed to 1.0; the other with the constant allowed to vary from one) would read:

If I_{cl} is expressed in $m^2 \cdot K \cdot W^{-1}$:

$$f_{cl} = 1.01 + 1.599 I_{cl} \qquad R^2 = 0.81 \qquad (13)$$

Or with a fixed constant:

$$f_{cl} = 1.00 + 1.697 I_{cl} \qquad R^2 = 0.98 \qquad (14)$$

If I_{cl} is expressed in clo:

$$f_{cl} = 1.01 + 0.248 I_{cl} \qquad R^2 = 0.81 \qquad (15)$$

Or with a fixed constant

$$f_{cl} = 1.00 + 0.263 I_{cl} \qquad R^2 = 0.98 \qquad (16)$$

The application range for which these new relations were tested is for I_{cl} between 0.2 and 1.0 clo.

It should be noted that the increase in R^2 when leaving out the constant is based on a different calculation of R^2 being used by statistical programs (e.g. SPSS) in models without (or fixed) constants. i.e. it should not be considered as an improvement in prediction. When the same calculation method is used as for models with constant, the R^2 value is the same as for without constant.

Dry Insulation

For operational reasons, we were unable to measure all ensembles in all three laboratories. As we observed good correlations between results in the different laboratories, the total and intrinsic whole-body clothing insulation values of the complete data set for the different ensembles measured at Loughborough University with the aforementioned correction to consider the effect of the different manikin shapes, chambers, and airflows were chosen to be presented in Tables 1 through 4.

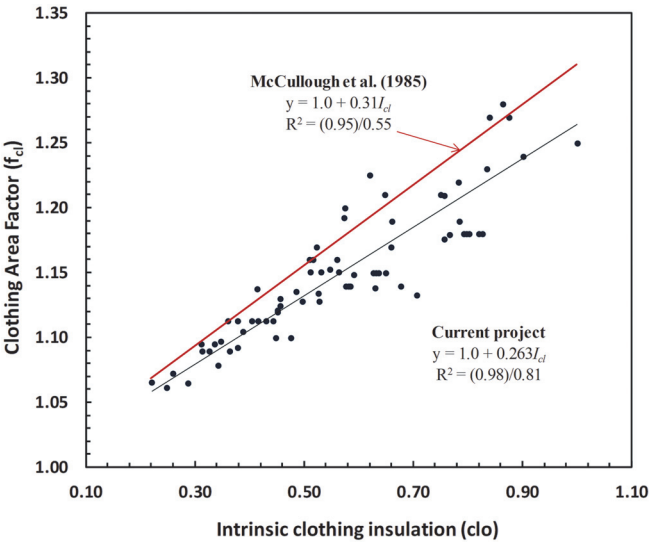


Figure 6 *Relationship between measured clothing area factor* f_{cl} *and intrinsic clothing insulation* I_{cl} *calculated using this* f_{cl} *value (see Equation 2). Compared are the data and regression line from the current project with the regression line obtained by McCullough et al. (1985). Explanation of two* R^2 *values in main text.*

Tables 1 and 2 contain dry insulation values in the standing posture for male and female ensembles, respectively. Tables 3 and 4 contain dry insulation values in the sitting posture for male and female ensembles, respectively. All measurements were conducted at an air velocity of 0.2 m/s^{-1} (0.67 ft/s^{-1}).

Moisture Permeability

The whole-body moisture permeability indices, as determined from the dry and evaporative resistance measurements conducted at CIOP-PIB Poland, of each of the 31 clothing ensembles tested are also presented in Tables 1 and 2. All values are the mean of duplicate or triplicate tests.

Postural Correction Factors

Mean correction coefficients for conversion from the standing to the sitting posture insulation are provided in Table 5. Separate values are provided for total insulation I_T and intrinsic insulation I_{cl}.

To calculate the sitting insulation from the standing insulation (Figure 1a), multiply the standing I_T or the I_{cl} values by the relevant correction factor provided in Table 5. It should be noted that this correction is for the posture change and the clothing compression. It does not include potential added insulation from the chair, which may be substantial depending

Table 1. Male Ensembles—Standing Posture (Air Velocity: 0.2 m/s^{-1} [0.66ft/s^{-1}])

Database Reference	I_T, m^2·K·W^{-1}	I_T, clo	SD I_T, %	I_a, m^2·K·W^{-1}	f_{cl}, n.d.	I_{cl}, m^2·K·W^{-1}	I_{cl}, (clo)	i_m, n.d.
Nude	0.098	0.63	0.30	0.098	1.00	0.000	0.00	—
ASTM	0.216	1.39	0.16	0.098	1.28	0.139	0.90	0.32
1	0.181	1.17	0.99	0.098	1.14	0.095	0.61	0.34
2	0.191	1.23	0.30	0.098	1.17	0.107	0.69	—
3	0.189	1.22	1.23	0.098	1.15	0.104	0.67	—
4	0.216	1.39	0.59	0.098	1.18	0.133	0.86	—
5	0.211	1.36	2.36	0.098	1.18	0.128	0.83	0.35
6	0.217	1.40	0.04	0.098	1.18	0.134	0.86	0.35
7	0.213	1.37	2.08	0.098	1.18	0.130	0.84	—
8	0.196	1.26	0.97	0.098	1.14	0.110	0.71	0.35
9	0.209	1.35	0.06	0.098	1.19	0.127	0.82	0.34
10	0.163	1.05	1.04	0.098	1.10	0.074	0.48	0.36
11	0.150	0.97	0.04	0.098	1.09	0.060	0.39	0.32
12	0.169	1.09	0.40	0.098	1.16	0.084	0.54	0.34
13	0.175	1.13	0.14	0.098	1.20	0.094	0.60	0.33
14	0.219	1.41	0.56	0.098	1.27	0.141	0.91	0.34
15	0.169	1.09	0.15	0.098	1.17	0.085	0.55	—
16	0.240	1.55	1.44	0.098	1.25	0.161	1.04	0.35
17	0.215	1.38	1.64	0.098	1.23	0.135	0.87	—
18	0.203	1.31	1.07	0.098	1.21	0.122	0.78	—
19	0.207	1.34	1.27	0.098	1.22	0.127	0.82	—
20	0.224	1.45	0.61	0.098	1.24	0.145	0.94	—
21	0.180	1.16	2.86	0.098	1.14	0.094	0.61	0.33
22	0.162	1.04	2.45	0.098	1.12	0.074	0.48	—
23	0.161	1.04	0.58	0.098	1.13	0.075	0.48	0.34
24	0.142	0.92	0.63	0.098	1.09	0.052	0.34	0.32
25	0.144	0.93	0.06	0.098	1.09	0.054	0.35	—
26	0.186	1.20	1.82	0.098	1.21	0.105	0.68	—
27	0.168	1.08	0.33	0.098	1.16	0.083	0.54	—
28	0.213	1.37	1.53	0.098	1.27	0.136	0.88	—
29	0.212	1.37	0.56	0.098	1.18	0.129	0.83	—
30	0.181	1.17	0.81	0.098	1.14	0.095	0.61	—

Note: Whole-body total clothing insulation I_T and standard deviation (%), intrinsic clothing insulation I_{cl}, clothing area factor f_{cl}, standard deviation (SD %), and vapor permeability index value i_m for the male clothing ensembles for standing posture at air velocity 0.2 m/s^{-1}. For detailed descriptions of the ensembles, refer to the appendix.

Table 2. Female Ensembles—Standing Posture (Air Velocity: 0.2 m/s^{-1} [0.66ft/s^{-1}])

Database Reference	I_T, m^2·K·W^{-1}	I_T, clo	SD I_T, %	I_a, m^2·K·W^{-1}	f_{cl}, n.d.	I_{cl}, m^2·K·W^{-1}	I_{cl}, clo	i_m, n.d.
Nude	0.098	0.63	0.30	0.098	1.00	0.000	0.00	—
ASTM	0.216	1.39	0.16	0.098	1.28	0.139	0.90	0.32
31	0.135	0.87	0.23	0.098	1.07	0.044	0.28	0.33
32	0.152	0.98	0.34	0.098	1.09	0.063	0.40	—
33	0.141	0.91	0.27	0.098	1.10	0.052	0.34	0.31
34	0.145	0.94	0.29	0.098	1.10	0.056	0.36	—
35	0.134	0.87	0.88	0.098	1.06	0.042	0.27	—
36	0.140	0.90	0.13	0.098	1.07	0.048	0.31	—
37	0.206	1.33	1.08	0.098	1.18	0.123	0.79	0.32
38	0.207	1.34	0.45	0.098	1.18	0.124	0.80	—
39	0.204	1.31	0.74	0.098	1.21	0.123	0.79	0.31
40	0.172	1.11	0.41	0.098	1.15	0.087	0.56	0.31
41	0.176	1.13	1.00	0.098	1.16	0.091	0.59	—
42	0.169	1.09	1.71	0.098	1.15	0.084	0.54	—
43	0.187	1.21	0.95	0.098	1.15	0.102	0.66	0.31
44	0.188	1.21	0.70	0.098	1.15	0.103	0.66	0.31
44b	0.190	1.22	1.11	0.098	1.19	0.107	0.69	—
45	0.191	1.23	0.94	0.098	1.15	0.106	0.68	—
46	0.177	1.14	0.62	0.098	1.15	0.092	0.59	—
47	0.154	1.00	0.19	0.098	1.14	0.068	0.44	0.34
48	0.182	1.17	0.02	0.098	1.15	0.096	0.62	0.32
49	0.148	0.95	0.24	0.098	1.11	0.060	0.39	0.30
50	0.162	1.04	0.06	0.098	1.12	0.075	0.48	—
51	0.151	0.97	0.30	0.098	1.11	0.063	0.40	0.30
52	0.155	1.00	0.90	0.098	1.11	0.067	0.43	—
53	0.156	1.01	0.26	0.098	1.11	0.068	0.44	—
54	0.161	1.04	1.06	0.098	1.11	0.073	0.47	0.31
55	0.172	1.11	0.57	0.098	1.13	0.086	0.55	—
56	0.130	0.84	0.46	0.098	1.07	0.038	0.24	—
57	0.147	0.95	0.56	0.098	1.10	0.058	0.37	—
58	0.173	1.12	0.16	0.098	1.13	0.086	0.56	0.33
59	0.201	1.30	0.65	0.098	1.13	0.115	0.74	0.34
60	0.166	1.07	0.26	0.098	1.14	0.080	0.51	—
61	0.174	1.13	0.20	0.098	1.15	0.089	0.58	—
62	0.181	1.17	0.89	0.098	1.23	0.101	0.65	—
63	0.168	1.09	0.55	0.098	1.13	0.081	0.53	0.30
64	0.189	1.22	1.04	0.098	1.14	0.103	0.66	—
65	0.159	1.02	0.93	0.098	1.11	0.071	0.46	—
66	0.153	0.98	0.77	0.098	1.11	0.064	0.41	—
67	0.167	1.08	0.31	0.098	1.10	0.078	0.50	—
68	0.162	1.04	0.50	0.098	1.12	0.074	0.48	0.30
69	0.148	0.95	0.73	0.098	1.08	0.057	0.37	—

Note: Whole-body total clothing insulation I_T and standard deviation (%), intrinsic clothing insulation I_{cl}, clothing area factor f_{cl}, standard deviation (SD %), and vapor permeability index value i_m for the female clothing ensembles for standing posture at air velocity 0.2 m/s^{-1}. For detailed descriptions of the ensembles, refer to the appendix.

Table 3. Male Ensembles—Sitting Posture (Air Velocity: 0.2 m/s^{-1} [0.66ft.s^{-1}])

Database Reference	I_T, m^2·K·W^{-1}	I_T, clo	SD I_T, %	I_a, m^2·K·W^{-1}	f_{cl}, n.d.	I_{cl}, m^2·K·W^{-1}	I_{cl}, clo	Sitting (0.2 m/s) Correction Factor, I_T, n.d.	Sitting (0.2 m/s) Correction Factor, I_{cl}, n.d.
Nude	0.108	0.70	0.09	0.108	1.00	0.000	0.00	1.11	—
ASTM	0.200	1.29	0.16	0.108	1.28	0.116	0.75	0.93	0.83
1	0.165	1.06	0.23	0.108	1.14	0.070	0.45	0.91	0.74
2	0.175	1.13	1.07	0.108	1.17	0.083	0.53	0.92	0.77
3	0.175	1.13	0.76	0.108	1.15	0.081	0.52	0.93	0.78
4	0.198	1.28	2.36	0.108	1.18	0.107	0.69	0.92	0.80
5	0.199	1.28	1.14	0.108	1.18	0.107	0.69	0.94	0.83
6	0.201	1.30	0.36	0.108	1.18	0.110	0.71	0.93	0.83
7	0.199	1.29	2.33	0.108	1.18	0.108	0.70	0.94	0.83
8	0.179	1.16	0.60	0.108	1.14	0.085	0.55	0.92	0.77
9	0.194	1.25	1.34	0.108	1.19	0.104	0.67	0.93	0.82
10	0.154	0.99	0.13	0.108	1.10	0.056	0.36	0.94	0.75
11	0.153	0.99	0.84	0.108	1.09	0.054	0.35	1.02	0.90
12	0.158	1.02	0.73	0.108	1.16	0.065	0.42	0.94	0.78
13	0.162	1.05	0.81	0.108	1.20	0.072	0.47	0.93	0.78
14	0.198	1.28	0.70	0.108	1.27	0.113	0.73	0.91	0.80
15	0.157	1.01	0.13	0.108	1.17	0.065	0.42	0.93	0.76
16	0.216	1.39	0.48	0.108	1.25	0.130	0.84	0.90	0.81
17	0.194	1.25	0.92	0.108	1.23	0.106	0.69	0.91	0.79
18	0.183	1.18	0.78	0.108	1.21	0.094	0.61	0.90	0.78
19	0.189	1.22	1.08	0.108	1.22	0.100	0.65	0.91	0.79
20	0.203	1.31	0.06	0.108	1.24	0.116	0.75	0.90	0.80
21	0.169	1.09	1.89	0.108	1.14	0.074	0.48	0.94	0.79
22	0.155	1.00	0.04	0.108	1.12	0.058	0.38	0.96	0.79
23	0.156	1.00	0.37	0.108	1.13	0.060	0.39	0.96	0.81
24	0.143	0.92	1.38	0.108	1.09	0.044	0.29	1.00	0.85
25	0.142	0.92	2.16	0.108	1.09	0.043	0.28	0.99	0.80
26	0.178	1.15	0.96	0.108	1.21	0.089	0.57	0.96	0.84
27	0.159	1.02	1.70	0.108	1.16	0.066	0.42	0.94	0.78
28	0.198	1.28	0.57	0.108	1.27	0.113	0.73	0.93	0.83
29	0.195	1.26	2.19	0.108	1.18	0.103	0.67	0.92	0.81
30	0.165	1.06	1.82	0.108	1.14	0.070	0.45	0.91	0.74

Note: Whole-body total clothing insulation I_T and standard deviation (SD %), intrinsic clothing insulation I_{cl}, clothing area factor f_{cl}, standard deviation (%) and moisture permeability index value i_m for the male clothing ensembles for sitting posture at air velocity 0.2 m/s^{-1}. Correction factors: $I_{sitting}$ = correction factor × $I_{standing}$. For detailed descriptions of the ensembles, refer to the appendix.

Table 4. Female Ensembles—Sitting Posture (Air Velocity: 0.2 m/s^{-1} [0.66 ft/s^{-1}])

Database Reference	I_T, m^2·K·W^{-1}	I_T, clo	SD I_T, %	I_a, m^2·K·W^{-1}	f_{cl}, n.d.	I_{cl}, m^2·K·W^{-1}	I_{cl}, clo	Sitting (0.2 m/s) Correction Factor, I_T, n.d.	Sitting (0.2 m/s) Correction Factor, I_{cl}, n.d.
Nude	0.108	0.70	0.09	0.108	1.00	0.00	0.00	1.11	—
ASTM	0.200	1.29	0.16	0.108	1.28	0.12	0.75	0.93	0.83
31	0.137	0.88	0.26	0.108	1.07	0.04	0.24	1.01	0.86
32	0.152	0.98	1.65	0.108	1.09	0.05	0.35	1.00	0.88
33	0.147	0.95	0.04	0.108	1.10	0.05	0.31	1.04	0.91
34	0.151	0.97	0.02	0.108	1.10	0.05	0.34	1.03	0.94
35	0.137	0.89	0.68	0.108	1.06	0.04	0.23	1.02	0.85
36	0.145	0.93	0.20	0.108	1.07	0.04	0.28	1.03	0.90
37	0.188	1.21	0.01	0.108	1.18	0.10	0.62	0.91	0.78
38	0.190	1.23	0.05	0.108	1.18	0.10	0.64	0.92	0.80
39	0.198	1.27	0.27	0.108	1.21	0.11	0.70	0.97	0.89
40	0.164	1.06	1.01	0.108	1.15	0.07	0.45	0.95	0.80
41	0.165	1.06	0.73	0.108	1.16	0.07	0.46	0.94	0.78
42	0.160	1.03	0.01	0.108	1.15	0.07	0.43	0.94	0.80
43	0.175	1.13	2.10	0.108	1.15	0.08	0.52	0.93	0.79
44	0.179	1.15	1.40	0.108	1.15	0.08	0.55	0.95	0.83
44b	0.177	1.14	0.96	0.108	1.19	0.09	0.56	0.93	0.81
45	0.177	1.14	0.52	0.108	1.15	0.08	0.53	0.93	0.78
46	0.163	1.05	0.27	0.108	1.15	0.07	0.45	0.92	0.76
47	0.147	0.95	0.42	0.108	1.14	0.05	0.33	0.95	0.75
48	0.166	1.07	0.59	0.108	1.15	0.07	0.47	0.91	0.76
49	0.147	0.95	1.13	0.108	1.11	0.05	0.32	1.00	0.82
50	0.159	1.03	0.14	0.108	1.12	0.06	0.41	0.99	0.85
51	0.150	0.97	1.98	0.108	1.11	0.05	0.34	1.00	0.85
52	0.157	1.01	2.42	0.108	1.11	0.06	0.39	1.01	0.91
53	0.157	1.01	1.05	0.108	1.11	0.06	0.39	1.00	0.89
54	0.158	1.02	0.07	0.108	1.11	0.06	0.39	0.98	0.83
55	0.169	1.09	1.41	0.108	1.13	0.07	0.48	0.98	0.87
56	0.133	0.86	0.23	0.108	1.07	0.03	0.21	1.02	0.88
57	0.146	0.94	0.14	0.108	1.10	0.05	0.31	0.99	0.84
58	0.162	1.05	0.01	0.108	1.13	0.07	0.43	0.94	0.77
59	0.189	1.22	0.15	0.108	1.13	0.09	0.60	0.94	0.81
60	0.153	0.99	0.78	0.108	1.14	0.06	0.37	0.93	0.73
61	0.164	1.06	0.18	0.108	1.15	0.07	0.46	0.94	0.79
62	0.170	1.10	1.02	0.108	1.23	0.08	0.53	0.94	0.82
63	0.159	1.02	0.25	0.108	1.13	0.06	0.41	0.94	0.77
64	0.174	1.12	0.03	0.108	1.14	0.08	0.51	0.92	0.77
65	0.161	1.04	1.01	0.108	1.11	0.06	0.41	1.02	0.89
66	0.157	1.01	0.72	0.108	1.11	0.06	0.38	1.03	0.93
67	0.171	1.10	0.22	0.108	1.10	0.07	0.47	1.02	0.94
68	0.166	1.07	0.31	0.108	1.12	0.07	0.45	1.03	0.94
69	0.150	0.97	0.22	0.108	1.08	0.05	0.32	1.02	0.86

Note: Whole-body total clothing insulation I_T and standard deviation (SD %), intrinsic clothing insulation I_{cl}, clothing area factor f_{cl}, standard deviation (%), and moisture permeability index value i_m for the female clothing ensembles for sitting posture at air velocity 0.2 m/s^{-1} (0.66 ft/s^{-1}). Correction factors: $I_{sitting}$ = correction factor × $I_{standing}$. For detailed descriptions of the ensembles, refer to the appendix.

Table 5. Correction Coefficients to Calculate Sitting Insulation from Standing Values (Air Velocity 0.2 m/s^{-1} [0.66 ft/s^{-1}])

	Male Total Insulation, I_T	Female Total Insulation, I_T	All Ensembles Mean Total Insulation, I_T	Male Intrinsic Insulation, I_{cl}	Female Intrinsic Insulation, I_{cl}	All Ensembles Mean Intrinsic Insulation, I_{cl}
Correction Coefficient Sitting, 0.2 m/s^{-1} (0.66 ft/s^{-1}), (n.d.)	0.93	0.97	0.96	0.80	0.84	0.82

*Correction factors calculated with nude manikin values excluded

on coverage and padding used. The corresponding surface air insulation I_a values for standing and seated postures were 0.098 m^2·K·W^{-1} and 0.108 m^2·K·W^{-1} (0.63 clo and 0.7 clo), respectively, i.e., intrinsic clothing insulation decreases when sitting down, but air insulation increases.

CONCLUSION

Three different manikins in three different laboratories were used to determine the clothing insulation values of 71 clothing configurations in both standing and sitting postures. Using different manikins in different research conditions to replicate these tests resulted in greater number of replications of each clothing ensemble measurement and provided insight into the inter-manikin variation. The similar results obtained from robust testing of the clothing ensembles at multiple research institutions enhances our confidence in the data obtained.

As expected, the data obtained from Sweden using the Tore manikin varied slightly to those obtained using the Newton manikin in Loughborough and Poland. The manikin shape is considered a contributor to this variation (Havenith et al. 2015), as this interacts with the clothing fit and drape, which subsequently affect the air layers in and on the clothing. However, the local conditions, mainly the airflow patterns, in the different climatic chambers may have also contributed to the variability observed. Though the values measured on the Polish and the U.K. manikins varied, the data from both "Newton" manikins, in chambers that were designed very differently with regards to airflow, are remarkably similar. Finally, the use of the manikin skin in the dry conditions may play a role. While its insulation was later subtracted from the data, its surface structure (textile surface versus smooth epoxy surface of Lund) may have affected convective flows.

Values observed for the vapor permeability index i_m for a static manikin in 0.2 m/s^{-1} air velocity were on average 0.33 ± 0.02 (n.d.). These data—relating to the static standing posture—provide an extensive updated database of Western clothing styles in different wear configurations. The current vapor permeability index values are lower, on average, than those reported previously for non-Western clothing ensembles (Havenith et al. 2015) and in earlier studies from ASHRAE (McCullough et al. 1989); however, it should be acknowledged that the non-Western data were derived from a different manikin ("Walter") using subtly different methods and using quite different clothing styles. In the earlier studies by McCullough and Hong (1994) clothing may have been on average looser or the style more voluminous, as suggested by the higher f_{cl} values in her data sets. This suggests bigger air layers potentially causing the higher i_m values.

UTILIZATION

This research relates to Chapter 9, "Thermal Comfort" of *ASHRAE Handbook—Fundamentals* (ASHRAE 2013b). The current research also relates to ASHRAE Standard 55 (2013a) and to ISO Standards 7730 (ISO 2005), 7933 (ISO 2004a), and 9920 (ISO 2009). The values for non-Western clothing insulation presented in this report will assist in the use of these standards and allow engineers to determine comfort guidelines for the built environment. Values for insulation and for the effects of posture, movement, and wind on insulation can be incorporated in the standards and in the ASHRAE Handbook.

Furthermore, data obtained on the insulation of individual body parts reported in the project report (Smallcombe et al 2020) can be used by modelers (e.g., CFD modeling) to incorporate realistic insulation data in their models.

These new data on Western clothing extend the current database and enhances the level of detail in terms of regional insulation distribution data for individual body segments (see report TR1760 for details [Smallcombe et al. 2020]). Additionally, it brings the available Western clothing insulation data in line with that available for non-Western clothing, for which detailed data were recently reported.

With regard to the application of these new data, it should be noted that where the validation of models has been based on specific choices for clothing insulation values (e.g. in ISO 7933's Predicted Heat Strain Index), a rerun of the validation may be required if different insulation values for the same clothing are used. However, for the clothing referenced in that standard, insulation tends to be low, and the impact of any changes potentially limited. For other standards, heat balance methods (ISO 7730 [2005], ASHRAE Standard 55 [2013a]), while outcomes may show slight changes with the more detailed information provided in this paper, the underlying method is not affected.

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NOMENCLATURE

f_{cl} = clothing area factor: ratio between the surface area of the clothed body (including unclothed parts) and the surface of the nude body, dimensionless

I_a = insulation of air layers surrounding manikin and clothing, $m^2 \cdot °C \cdot W^{-1}$ ($m^2 \cdot K \cdot W^{-1}$) (clo)

I_{cl} = intrinsic insulation value (without surrounding surface boundary air layers), $m^2 \cdot °C \cdot W^{-1}$ ($m^2 \cdot K \cdot W^{-1}$) (clo)

i_m = moisture permeability index (Woodcock permeability index), ratio between the total dry-clothing heat resistance I_T and the total evaporative clothing heat resistance $R_{e,T}$ for a clothing ensemble, divided by the Lewis relation, (dimensionless)

I_T = total insulation value of clothing plus surrounding boundary air layers, $m^2 \cdot K \cdot W^{-1}$ (clo)

L = Lewis relation, ratio of the evaporative heat transfer coefficient to the convective heat transfer coefficient, 16.5 K/kPa

n.d. = nondimensional number

$R_{e,a}$ = *evaporative resistance of the boundary air layer. Resistance to vapor transport for the whole body at the boundary (skin or clothing), $m^2 \cdot Pa \cdot W^{-1}$ ($1\ m^2 \cdot Pa \cdot W^{-1} = 0.0659\ lbf \cdot h \cdot Btu^{-1}$)*

$R_{e,T}$ *Evaporative resistance of a clothing ensemble, resistance to vapor transport of a uniform layer of insulation covering the entire body that has the same effect of evaporative heat loss as the actual clothing under the tested conditions, $m^2.Pa.W^{-1}$ ($1\ m^2 \cdot Pa \cdot W^{-1} = 0.0659\ lbf \cdot h \cdot Btu^{-1}$)*

APPENDIX

Descriptions of Clothing Ensembles

Descriptions and photographs of the male, female, and reference clothing ensembles tested are provided in the following table. The weight of each clothing garment was measured to the nearest 1.0 g (0.035 oz) using a high precision digital scale (Metter Toledo kcc150, Metter Toledo, Leicester, U.K.).

The material thickness of each garment was determined to the nearest 0.01 mm (0.0004 in.) using a digital, dead-weight thickness gauge (Schmidt control instruments) at a measuring pressure of 1 kPa. The thickness gauge used was appropriate for use with woven, knitted, and textured fabrics. Thickness measurements were conducted according to ASTM Test Method D1777 (2019). For each clothing garment, thickness measurements were conducted in triplicate, with the median value reported.

Ref.	Garment	Weight	Thickness	Material
	ASTM (Reference)			
	Boxer shorts	88 g	0.45 mm	CO
	Socks (sport)	53 g	0.93 mm	CO, PA, ELA
	T-shirt	158 g	0.64 mm	CO
	ASTM shirt	430 g	0.69 mm	ARA, KEV
	ASTM pants	442 g	0.63 mm	ARA, KEV
	Trainers/ sneakers	424 g	—	PES, RB
	Shirt not tucked into (over) pants. Shirt top button open.			
	1			
	Boxer shorts	88 g	0.45 mm	CO
	Socks	44 g	0.93 mm	CO
	Office Shirt	249 g	0.29 mm	PES, CO
	Suit pants	325 g	0.91 mm	PES, VIS
	Shoes	812 g	—	LTH, RB
	Shirt tucked into pants. Shirt top button open.			
	2			
	Boxer shorts	88 g	0.45 mm	CO
	Socks	44 g	0.93 mm	CO
	T-shirt	158 g	0.64 mm	CO
	Office shirt	249 g	0.29 mm	PES, CO
	Suit pants	325 g	0.91 mm	PES, VIS
	Shoes	812 g	—	LTH, RB
	T-shirt tucked into pants. Shirt top button open.			
	3			
	Boxer shorts	88 g	0.45 mm	CO
	Socks	44 g	0.93 mm	CO
	Office shirt	249 g	0.29 mm	PES, CO
	Suit pants	325 g	0.91 mm	PES, VIS
	Waistcoat	855 g	1.28 mm (front) 0.21 mm (back)	PES, VIS
	Tie	40 g	3.10 mm	PES
	Shoes	812 g	—	LTH, RB
	Shirt tucked into pants. Waistcoat buttons closed.			
	4			
	Boxer shorts	88 g	0.45 mm	CO
	Socks	44 g	0.93 mm	CO
	Office shirt	249 g	0.29 mm	PES, CO
	Suit pants	325 g	0.91 mm	PES, VIS
	Tweed blazer jacket	855 g	1.56 mm	PES, WO, PA,
	Tie	40 g	3.10 mm	PES
	Shoes	812 g	—	LTH, RB
	Shirt tucked into pants. Waistcoat buttons closed. Jacket buttons closed.			
	5			
	Boxer shorts	88 g	0.45 mm	CO
	Socks	44 g	0.93 mm	CO
	Office shirt	249 g	0.29 mm	PES, CO
	Suit pants	325 g	0.91 mm	PES, VIS
	Suit Jacket	630 g	1.84 mm (front) 0.54 mm (back)	PES, VIS
	Tie	40 g	3.10 mm	PES
	Shoes	812 g	—	LTH, RB
	Shirt tucked into pants. Suit jacket button closed.			

Key:ARA, Aramid; CO, Cotton; ELA, Elastane; KEV, Kevlar; LTH, Leather; NYL, Nylon; PA, Polyamide; PES, Polyester; RB, Rubber; VIS, Viscose

Ref.	Garment	Weight	Thickness	Material
6				
	Boxer shorts	88 g	0.45 mm	CO
	Socks	44 g	0.93 mm	CO
	T-shirt	158 g	0.64 mm	CO
	Office Shirt	249 g	0.29 mm	PES, CO
	Suit pants	325 g	0.91 mm	PES, VIS
	Suit Jacket	630 g	1.84 mm (front) 0.54 mm (back)	PES, VIS
	Tie	40 g	3.10 mm	PES
	Shoes	812 g	—	LTH, RB
	Shirt tucked into pants. Suit jacket button closed.			
7				
	Boxer shorts	88 g	0.45 mm	CO
	Socks	44 g	0.93 mm	CO
	Office Shirt	249 g	0.29 mm	PES, CO
	Suit pants	325 g	0.91 mm	PES, VIS
	Waistcoat	855 g	1.28 mm (front) 0.21 mm (back)	PES, VIS
	Suit Jacket	630 g	1.84 mm (front) 0.54 mm (back)	PES, VIS
	Tie	40 g	3.10 mm	PES
	Shoes	812 g	—	LTH, RB
	Shirt tucked into pants. Suit jacket button closed.			
8				
	Boxer shorts	88 g	0.45 mm	CO
	Socks	44 g	0.93 mm	CO
	Office Shirt	249 g	0.29 mm	PES, CO
	Suit pants	325 g	0.91 mm	PES, VIS
	V-neck sweater	267 g	1.07 mm	CO, NYL
	Shoes	812 g	—	LTH, RB
	Shirt tucked into pants. Shirt top button open.			

Ref.	Garment	Weight	Thickness	Material
9				
	Boxer shorts	88 g	0.45 mm	CO
	Socks	44 g	0.93 mm	CO
	Office Shirt	249 g	0.29 mm	PES, CO
	Suit pants	325 g	0.91 mm	PES, VIS
	Harrington Jacket	820 g	1.11 mm	CO PES
	Tie	40 g	3.10 mm	PES
	Shoes	812 g	—	LTH, RB
	Shirt tucked into pants. Shirt top button closed. Harrington jacket zipped up.			
10				
	Boxer shorts	88 g	0.45 mm	CO
	Socks	44 g	0.93 mm	CO
	Polo shirt	233 g	0.84 mm	CO, ELA
	Suit pants	325 g	0.91 mm	PES, VIS
	Shoes	812 g	—	LTH, RB
	Polo shirt tucked into pants. Polo shirt top button open.			
11				
	Boxer shorts	88 g	0.45 mm	CO
	Socks	44 g	0.93 mm	CO
	Polo shirt	233 g	0.84 mm	CO, ELA
	Jeans (skinny fit)	544 g	0.95 mm	CO, ELA
	Shoes	812 g	—	LTH, RB
	Polo shirt tucked into jeans. Polo shirt top button open.			

Key:ARA, Aramid; CO, Cotton; ELA, Elastane; KEV, Kevlar; LTH, Leather; NYL, Nylon; PA, Polyamide; PES, Polyester; RB, Rubber; VIS, Viscose

Ref.	Garment	Weight	Thickness	Material
12				
	Boxer shorts	88 g	0.45 mm	CO
	Socks	44 g	0.93 mm	CO
	Polo shirt	233 g	0.84 mm	CO, ELA
	Jeans (straight fit)	608 g	0.91 mm	CO, ELA
	Shoes	812 g	—	LTH, RB
	Polo shirt tucked into jeans. Polo shirt top button open.			
13				
	Boxer shorts	88 g	0.45 mm	CO
	Socks	44 g	0.93 mm	CO
	Polo shirt	233 g	0.84 mm	CO, ELA
	Jeans (loose fit)	636 g	0.91 mm	CO, ELA
	Shoes	812 g	—	LTH, RB
	Polo shirt tucked into jeans. Polo shirt top button open.			
14				
	Boxer shorts	88 g	0.45 mm	CO
	Socks	44 g	0.93 mm	CO
	Polo shirt	233 g	0.84 mm	CO, ELA
	Jeans (straight fit)	608 g	0.91 mm	CO, ELA
	Tweed blazer jacket	855 g	1.56 mm	PES, WO. PA
	Shoes	812 g	—	LTH, RB
	Polo shirt tucked into jeans. Polo shirt top button open. Jacket button open.			

Ref.	Garment	Weight	Thickness	Material
15				
	Boxer shorts	88 g	0.45 mm	CO
	Socks	44 g	0.93 mm	CO
	T-shirt	158 g	0.64 mm	CO
	Jeans (straight fit)	608 g	0.91 mm	CO, ELA
	Shoes	812 g	—	LTH, RB
	T-shirt not tucked in (over) jeans.			
16				
	Boxer shorts	88 g	0.45 mm	CO
	Socks	44 g	0.93 mm	CO
	T-shirt	158 g	0.64 mm	CO
	Jeans (straight fit)	608 g	0.91 mm	CO, ELA
	Thick hooded jacket (“Hoodie”)	830 g	4.65 mm	CO, PES
	Shoes	812 g	—	LTH, RB
	T-shirt tucked into jeans. Hood draped on back.			
17				
	Boxer shorts	88 g	0.45 mm	CO
	Socks	44 g	0.93 mm	CO
	T-shirt	158 g	0.64 mm	CO
	Jeans (straight fit)	608 g	0.91 mm	CO, ELA
	Thin hooded jacket (“Hoodie”)	540 g	1.96 mm	CO, PES
	Shoes	812 g	—	LTH, RB
	T-shirt tucked into jeans. Hood draped on back.			

Key:ARA, Aramid; CO, Cotton; ELA, Elastane; KEV, Kevlar; LTH, Leather; NYL, Nylon; PA, Polyamide; PES, Polyester; RB, Rubber; VIS, Viscose

Ref.	Garment	Weight	Thickness	Material
18				
	Boxer shorts	88 g	0.45 mm	CO
	Socks	44 g	0.93 mm	CO
	T-shirt	158 g	0.64 mm	CO
	Check cotton shirt (thick)	279 g	0.71 mm	CO
	Jeans (straight fit)	608 g	0.91 mm	CO, ELA
	Shoes	812 g	—	LTH, RB
	T-shirt and shirt tucked into jeans. Shirt top button open.			
19				
	Boxer shorts	88 g	0.45 mm	CO
	Socks	44 g	0.93 mm	CO
	T-shirt	158 g	0.64 mm	CO
	Check cotton shirt (thick)	279 g	0.71 mm	CO
	Jeans (straight fit)	608 g	0.91 mm	CO, ELA
	Shoes	812 g	—	LTH, RB
	Identical to ensemble 18 but with shirt over jeans (untucked).			
20				
	Boxer shorts	88 g	0.45 mm	CO
	Socks	44 g	0.93 mm	CO
	T-shirt	158 g	0.64 mm	CO
	Check cotton shirt (thick)	279 g	0.71 mm	CO
	Thin hooded jacket ("Hoodie")	540 g	1.96 mm	CO, PES
	Jeans (straight fit)	608 g	0.91 mm	CO, ELA
	Shoes	812 g	—	LTH, RB
	T-shirt and shirt tucked into jeans. Hood draped on back.			

Ref.	Garment	Weight	Thickness	Material
21				
	Boxer shorts	88 g	0.45 mm	CO
	Socks	44 g	0.93 mm	CO
	Office shirt	249 g	0.29 mm	PES, CO
	Chino pants	422 g	0.70 mm	CO, ELA
	Shoes	812 g	—	LTH, RB
	Shirt tucked into chino pants. Shirt top button open.			
22				
	Boxer shorts	88 g	0.45 mm	CO
	Socks	44 g	0.93 mm	CO
	Short-sleeve shirt	200 g	0.29 mm	LIN, CO
	Chino pants	422 g	0.70 mm	CO, ELA
	Shoes	812 g	—	LTH, RB
	Shirt tucked into pants. Shirt top button open.			
23				
	Boxer shorts	88 g	0.45 mm	CO
	Socks	44 g	0.93 mm	CO
	T-shirt	158 g	0.64 mm	CO
	Chino pants	422 g	0.70 mm	CO, ELA
	Trainers/ sneakers	424 g	—	PES, RB
	T-shirt not tucked in (over) chino pants.			

Key:ARA, Aramid; CO, Cotton; ELA, Elastane; KEV, Kevlar; LTH, Leather; NYL, Nylon; PA, Polyamide; PES, Polyester; RB, Rubber; VIS, Viscose

Ref.	Garment	Weight	Thickness	Material
	24			
	Boxer shorts	88 g	0.45 mm	CO
	Socks	44 g	0.93 mm	CO
	Polo shirt	233 g	0.84 mm	CO, ELA
	Chino shorts	265 g	0.56 mm	CO
	Shoes	812 g	—	LTH, RB
	Polo shirt tucked into shorts. Polo shirt top button open.			
	25			
	Boxer shorts	88 g	0.45 mm	CO
	Socks	44 g	0.93 mm	CO
	T-shirt	158 g	0.64 mm	CO
	Chino shorts	265 g	0.56 mm	CO
	Shoes	812 g	—	LTH, RB
	T-shirt not tucked into (over) shorts.			
	26 (Kansas State Uniform)			
	Boxer shorts	88 g	0.45 mm	CO
	Socks	44 g	0.93 mm	CO
	KSU shirt	300 g		
	KSU pants	517 g		
	Trainers/ sneakers	424 g	—	PES, RB
	Shirt not tucked into (over) pants.			

Ref.	Garment	Weight	Thickness	Material
	27			
	Briefs	42 g	0.65 mm	CO
	Socks	44 g	0.93 mm	CO
	Polo shirt	233 g	0.84 mm	CO, ELA
	Jeans (straight fit)	608 g	0.91 mm	CO, ELA
	Shoes	812 g	—	LTH, RB
	Polo shirt tucked into jeans. Polo shirt top button open.			
	Same as 12 but with briefs.			
	28			
	Briefs	42 g	0.65 mm	CO
	Socks	44 g	0.93 mm	CO
	Polo shirt	233 g	0.84 mm	CO, ELA
	Jeans (straight fit)	608 g	0.91 mm	CO, ELA
	Tweed blazer jacket	855 g	1.56 mm	PES, WO. PA
	Shoes	812 g	—	LTH, RB
	Polo shirt tucked into jeans. Polo shirt top button open. Jacket button open.			
	Same as 14 but with briefs.			
	29			
	Briefs	42 g	0.65 mm	CO
	Socks	44 g	0.93 mm	CO
	Office shirt	249 g	0.29 mm	PES, CO
	Suit pants	325 g	0.91 mm	PES, VIS
	Suit Jacket	630 g	1.84 mm (front) 0.54 mm (back)	PES, VIS
	Tie	40 g	3.10 mm	PES
	Shoes	812 g	—	LTH, RN
	Shirt tucked into pants. Suit jacket button closed. Same as 5 but with briefs.			

Key:ARA, Aramid; CO, Cotton; ELA, Elastane; KEV, Kevlar; LTH, Leather; NYL, Nylon; PA, Polyamide; PES, Polyester; RB, Rubber; VIS, Viscose

Ref.	Garment	Weight	Thickness	Material
	30			
	Briefs	42 g	0.65 mm	CO
	Socks	44 g	0.93 mm	CO
	Office Shirt	249 g	0.29 mm	PES, CO
	Suit pants	325 g	0.91 mm	PES, VIS
	Shoes	812 g	—	LTH, RB
	Shirt tucked into pants. Shirt top button open. Same as 1 but with briefs.			
	31			
	Bikini style briefs	29 g	0.65 mm	CO
	Non-wired bra	62 g	0.70 mm	PES, PA
	Sleeveless print dress	333 g	0.52 mm	PES
	Shoes	593 g	—	LTH, RB
	32			
	Bikini style briefs	29 g	0.65 mm	CO
	Non-wired bra	62 g	0.70 mm	PES, PA
	Sleeveless print dress	333 g	0.52 mm	PES
	Cardigan	210 g	1.04 mm	CO, NYL, VIS
	Shoes	593 g	—	LTH, RB
	Cardigan buttons closed.			

Ref.	Garment	Weight	Thickness	Material
	33			
	Bikini style briefs	29 g	0.65 mm	CO
	Non-wired bra	62 g	0.70 mm	PES, PA
	Short-sleeve jacquard dress	448 g	1.24 mm	PES, VIS. ELA
	Shoes	593 g	—	LTH, RB
	34			
	Bikini style briefs	29 g	0.65 mm	CO
	Non-wired bra	62 g	0.70 mm	PES, PA
	Short-sleeve jacquard dress	448 g	1.24 mm	PES, VIS. ELA
	Tights (10 denier)	17 g	n/a	NYL, ELA
	Shoes	593 g	—	LTH, RB
	35			
	Bikini style briefs	29 g	0.65 mm	CO
	Non-wired bra	62 g	0.70 mm	PES, PA
	Short sleeve “fit and flare” dress	227 g	0.31 mm	VIS
	Sandals	593 g	—	PES

Key:ARA, Aramid; CO, Cotton; ELA, Elastane; KEV, Kevlar; LTH, Leather; NYL, Nylon; PA, Polyamide; PES, Polyester; RB, Rubber; VIS, Viscose

Ref.	Garment	Weight	Thickness	Material
	36			
	Bikini style briefs	29 g	0.65 mm	CO
	Non-wired bra	62 g	0.70 mm	PES, PA
	Long-sleeve “fit and flare’ print dress	258 g	0.34 mm	PES
	Sandals	593 g	—	PES
	37			
	Bikini style briefs	29 g	0.65 mm	CO
	Non-wired bra	62 g	0.70 mm	PES, PA
	Lace-sleeve blouse	190 g	0.39 mm (back) 0.69 (side)	VIS NYL
	Suit pants (tapered)	459 g	0.80 mm	PES, VIS ELA
	Suit jacket	742 g	1.60 mm	PES, VIS, ELA
	Shoes	593 g	—	LTH, RB
	Blouse tucked into pants. Jacket buttons closed.			
	38			
	Bikini style briefs	29 g	0.65 mm	CO
	Non-wired bra	62 g	0.70 mm	PES, PA
	Lace-sleeve blouse	190 g	0.39 mm (back) 0.69 (side)	VIS NYL
	Pencil skirt	373 g	0.75 mm	PES, VIS, ELA
	Suit jacket	742 g	1.60 mm	PES, VIS, ELA
	Shoes	593 g	—	LTH, RB
	Blouse tucked into skirt. Jacket buttons closed.			

Ref.	Garment	Weight	Thickness	Material
	39			
	Bikini style briefs	29 g	0.65 mm	CO
	Non-wired bra	62 g	0.70 mm	PES, PA
	Lace-sleeve blouse	190 g	0.39 mm (back) 0.69 (side)	VIS NYL
	Pencil skirt	373 g	0.75 mm	PES, VIS, ELA
	Suit jacket	742 g	1.60 mm	PES, VIS, ELA
	Tights (10 denier)	17 g	—	NYL ELA
	Shoes	593 g	—	LTH, RB
	Blouse tucked into skirt. Jacket button closed.			
	40			
	Bikini style briefs	29 g	0.65 mm	CO
	Non-wired bra	62 g	0.70 mm	PES, PA
	Long-sleeve blouse	190 g	0.31 mm	PES
	Pencil skirt	373 g	0.75 mm	PES, VIS, ELA
	Shoes	593 g	—	LTH, RB
	Blouse tucked into skirt.			
	41			
	Bikini style briefs	29 g	0.65 mm	CO
	Non-wired bra	62 g	0.70 mm	PES, PA
	Long-sleeve blouse	190 g	0.31 mm	PES
	Pencil skirt	373 g	0.75 mm	PES, VIS, ELA
	Shoes	593 g	—	LTH, RB
	Same as ensemble 40 but blouse over the pencil skirt (untucked).			

Ref.	Garment	Weight	Thickness	Material
42				
	Bikini style briefs	29 g	0.65 mm	CO
	Long-sleeve blouse	190 g	0.31 mm	PES
	Pencil skirt	373 g	0.75 mm	PES, VIS, ELA
	Shoes	593 g	—	LTH, RB
	Same as ensemble 40 without bra.			
43				
	Bikini style briefs	29 g	0.65 mm	CO
	Non-wired bra	62 g	0.70 mm	PES, PA
	Long-sleeve blouse	190 g	0.31 mm	PES
	Pencil skirt	373 g	0.75 mm	PES, VIS, ELA
	Cardigan	210 g	1.04 mm	CO, NYL, ELA
	Shoes	593 g	—	LTH, RB
	Blouse tucked into skirt. Cardigan buttons closed.			
44				
	Bikini style briefs	29 g	0.65 mm	CO
	Non-wired bra	62 g	0.70 mm	PES, PA
	Long-sleeve blouse	190 g	0.31 mm	PES
	Pencil skirt	373 g	0.75 mm	PES, VIS, ELA
	Cardigan	210 g	1.04 mm	CO, NYL, ELA
	Tights (10 denier)	17 g	—	NYL ELA
	Shoes	593 g	—	LTH, RB
	Blouse tucked into skirt. Cardigan buttons closed.			

Ref.	Garment	Weight	Thickness	Material
45				
	Bikini style briefs	29 g	0.65 mm	CO
	Non-wired bra	62 g	0.70 mm	PES, PA
	Strap vest top	94 g	0.65 mm	CO, VIS, ELA
	Long-sleeve blouse	190 g	0.31 mm	PES
	Pencil skirt	373 g	0.75 mm	PES, VIS, ELA
	Cardigan	210 g	1.04 mm	CO, NYL, ELA
	Shoes	593 g	—	LTH, RB
	Vest top underneath blouse. Blouse tucked into skirt. Cardigan buttons closed.			
46				
	Bikini style briefs	29 g	0.65 mm	CO
	Non-wired bra	62 g	0.70 mm	PES, PA
	Strap vest top	94 g	0.65 mm	CO, VIS, ELA
	Long-sleeve blouse	190 g	0.31 mm	PES
	Pencil skirt	373 g	0.75 mm	PES, VIS, ELA
	Shoes	593 g	—	LTH, RB
	Vest top underneath blouse. Blouse tucked into skirt.			
47				
	Bikini style briefs	29 g	0.65 mm	CO
	Non-wired bra	62 g	0.70 mm	PES, PA
	Sleeveless blouse	195 g	0.34 mm	PES, ELA
	Pencil skirt	373 g	0.75 mm	PES, VIS, ELA
	Shoes	593 g	—	LTH, RB
	Blouse tucked into skirt.			

Key: ARA, Aramid; CO, Cotton; ELA, Elastane; KEV, Kevlar; LTH, Leather; NYL, Nylon; PA, Polyamide; PES, Polyester; RB, Rubber; VIS, Viscose

Ref.	Garment	Weight	Thickness	Material
48				
	Bikini style briefs	29 g	0.65 mm	CO
	Non-wired bra	62 g	0.70 mm	PES, PA
	Sleeveless blouse	195 g	0.34 mm	PES, ELA
	Cardigan	210 g	1.04 mm	CO, NYL, ELA
	Pencil skirt	373 g	0.75 mm	PES, VIS, ELA
	Shoes	593 g	—	LTH, RB
Blouse tucked into skirt. Cardigan buttons open.				
49				
	Bikini style briefs	29 g	0.65 mm	CO
	Non-wired bra	62 g	0.70 mm	PES, PA
	Long-sleeve blouse	190 g	0.31 mm	PES
	Mini skirt (thick)	312 g	1.58 mm	CO
	Shoes	593 g	—	LTH, RB
Blouse tucked into skirt.				
50				
	Bikini style briefs	29 g	0.65 mm	CO
	Non-wired bra	62 g	0.70 mm	PES, PA
	Long-sleeve blouse	190 g	0.31 mm	PES
	Cardigan	210 g	1.04 mm	CO, NYL, ELA
	Mini skirt	312 g	1.58 mm	CO
	Tights (10 denier)	17 g	—	NYL ELA
	Shoes	593 g	—	LTH, RB
Blouse tucked into skirt. Cardigan buttons open.				

Ref.	Garment	Weight	Thickness	Material
51				
	Bikini style briefs	29 g	0.65 mm	CO
	Non-wired bra	62 g	0.70 mm	PES, PA
	Long-sleeve blouse	190 g	0.31 mm	PES
	Mini skirt (thick)	312 g	1.58 mm	CO
	Tights (*10 denier*)	17 g	—	NYL ELA
	Shoes	593 g	—	LTH, RB
Blouse tucked into skirt.				
52				
	Bikini style briefs	29 g	0.65 mm	CO
	Non-wired bra	62 g	0.70 mm	PES, PA
	Long-sleeve blouse	190 g	0.31 mm	PES
	Mini skirt (thick)	312 g	1.58 mm	CO
	Tights (*80 denier*)	53 g	—	NYL ELA
	Shoes	593 g	—	LTH, RB
Blouse tucked into skirt.				
53				
	Bikini style briefs	29 g	0.65 mm	CO
	Non-wired bra	62 g	0.70 mm	PES, PA
	Long-sleeve blouse	190 g	0.31 mm	PES
	Mini skirt (thick)	312 g	1.58 mm	CO
	Tights (*200 denier*)	124 g	—	NYL ELA
	Shoes	593 g	—	LTH, RB
Blouse tucked into skirt.				

KEY:ARA, Aramid; CO, Cotton; ELA, Elastane; KEV, Kevlar; LTH, Leather; NYL, Nylon; PA, Polyamide; PES, Polyester; RB, Rubber; VIS, Viscose

Ref.	Garment	Weight	Thickness	Material
54				
	Bikini style briefs	29 g	0.65 mm	CO
	Non-wired bra	62 g	0.70 mm	PES, PA
	Long-sleeve blouse	190 g	0.31 mm	PES
	Mini skirt (thick)	312 g	1.58 mm	CO
	Leggings	185 g	—	NYL ELA
	Shoes	593 g	—	LTH, RB
Blouse tucked into skirt.				
55				
	Bikini style briefs	29 g	0.65 mm	CO
	Non-wired bra	62 g	0.70 mm	PES, PA
	Long-sleeve blouse	190 g	0.31 mm	PES
	Mini skirt (thick)	312 g	1.58 mm	CO
	Knee-high boots	977 g	—	LTH, RB
Blouse tucked into skirt.				
56				
	Bikini style briefs	29 g	0.65 mm	CO
	Non-wired bra	62 g	0.70 mm	PES, PA
	Cap-sleeve top	152 g	0.57 mm	CO
	Short mini skirt (thin)	203 g	0.60 mm	PES, VIS, ELA
	Shoes	593 g	—	LTH, RB
Cap-sleeve top tucked into skirt.				

Ref.	Garment	Weight	Thickness	Material
57				
	Bikini style briefs	29 g	0.65 mm	CO
	Non-wired bra	62 g	0.70 mm	PES, PA
	Long-sleeve blouse	190 g	0.31 mm	PES
	Denim Midi Skirt	290 g	0.92 mm	CO, PES, ELA
	Shoes	593 g	—	LTH, RB
Blouse tucked into skirt.				
58				
	Bikini style briefs	29 g	0.65 mm	CO
	Non-wired bra	62 g	0.70 mm	PES, PA
	Long-sleeve blouse	190 g	0.31 mm	PES
	Suit pants (tapered)	459 g	0.80 mm	PES, VIS ELA
	Shoes	593 g	—	LTH, RB
Blouse tucked into pants.				
59				
	Bikini style briefs	29 g	0.65 mm	CO
	Non-wired bra	62 g	0.70 mm	PES, PA
	Long-sleeve blouse	190 g	0.31 mm	PES
	Long-sleeve jumper	302 g	2.38 mm	ACRY
	Suit pants (tapered)	459 g	0.80 mm	PES, VIS ELA
	Shoes	593 g	—	LTH, RB
Blouse tucked into pants.				

Key: ARA, Aramid; Co, Cotton; Ela, Elastane; Kev, Kevlar; Lth, Leather; Nyl, Nylon; Pa, Polyamide; Pes, Polyester; Rb, Rubber; Vis, Viscose.

Ref.	Garment	Weight	Thickness	Material	Ref.	Garment	Weight	Thickness	Material
60					**63**				
	Bikini style briefs	29 g	0.65 mm	CO		Bikini style briefs	29 g	0.65 mm	CO
	Non-wired bra	62 g	0.70 mm	PES, PA		Non-wired bra	62 g	0.70 mm	PES, PA
	Lace-sleeve blouse	190 g	0.39 mm (back) 0.69 (side)	VIS NYL		Long-sleeve blouse	190 g	0.31 mm	PES
	Suit pants (*tapered fit*)	459 g	0.80 mm	PES, VIS ELA		Capri pants (*cropped*)	364 g	0.64 mm	CO, ELA
	Shoes	593 g	—	LTH, RB		Shoes	593 g	—	LTH, RB
	Blouse tucked into pants.					Blouse tucked into pants.			
61					**64**				
	Bikini style briefs	29 g	0.65 mm	CO		Bikini style briefs	29 g	0.65 mm	CO
	Non-wired bra	62 g	0.70 mm	PES, PA		Non-wired bra	62 g	0.70 mm	PES, PA
	Long-sleeve blouse	190 g	0.31 mm	PES		Strap vest top	94 g	0.65 mm	CO, VIS, ELA
	Suit pants (*bootcut fit*)	499 g	0.80 mm	PES, VIS ELA		Long-sleeve jumper	302 g	2.38 mm	ACRY
	Shoes	593 g	—	LTH, RB		Capri pants (*cropped*)	364 g	0.64 mm	CO, ELA
	Blouse tucked into pants.					Shoes	593 g	—	LTH, RB
62					**65**				
	Bikini style briefs	29 g	0.65 mm	CO		Bikini style briefs	29 g	0.65 mm	CO
	Non-wired bra	62 g	0.70 mm	PES, PA		Non-wired bra	62 g	0.70 mm	PES, PA
	Long-sleeve blouse	190 g	0.31 mm	PES		Long-sleeve blouse	190 g	0.31 mm	PES
	Suit pants (*loose fit*)	530 g	0.80 mm	PES, VIS ELA		Ripped Jeans	491 g	0.94 mm	CO, ELA
	Shoes	593 g	—	LTH, RB		Shoes	593 g	—	LTH, RB
	Blouse tucked into pants.					Blouse not tucked into (over) jeans.			

Key: ARA, Aramid; CO, Cotton; ELA, Elastane; KEV, Kevlar; LTH, Leather; NYL, Nylon; PA, Polyamide; PES, Polyester; RB, Rubber; VIS, Viscose.

Ref.	Garment	Weight	Thickness	Material
	66			
	Bikini style briefs	29 g	0.65 mm	CO
	Non-wired bra	62 g	0.70 mm	PES, PA
	Long-sleeve blouse	190 g	0.31 mm	PES
	Jeans (skinny fit)	544 g	0.95 mm	CO, ELA
	Shoes	593 g	—	LTH, RB
	Blouse not tucked into (over) jeans.			
	67			
	Bikini style briefs	29 g	0.65 mm	CO
	Non-wired bra	62 g	0.70 mm	PES, PA
	Long-sleeve blouse	190 g	0.31 mm	PES
	Jeans (skinny fit)	544 g	0.95 mm	CO, ELA
	Knee-high boots	977 g	—	LTH, RB
	Blouse tucked into jeans. Boots over jeans.			
	68			
	Bikini style briefs	29 g	0.65 mm	CO
	Non-wired bra	62 g	0.70 mm	PES, PA
	Long-sleeve tunic top	211 g	0.57 mm	PES, VIS
	Jeans (skinny fit)	544 g	0.95 mm	CO, ELA
	Knee-high boots	977 g	—	LTH, RB
	Tunic top not tucked in (over) jeans. Boots over jeans.			

Ref.	Garment	Weight	Thickness	Material
	69			
	Bikini style briefs	29 g	0.65 mm	CO
	Non-wired bra	62 g	0.70 mm	PES, PA
	Long-sleeve tunic top	211 g	0.57 mm	PES, VIS
	Leggings	185 g	—	NYL ELA
	Shoes	593 g	—	LTH, RB
	Tunic top over leggings.			

Key:ARA, Aramid; CO, Cotton; ELA, Elastane; KEV, Kevlar; LTH, Leather; NYL, Nylon; PA, Polyamide; PES, Polyester; RB, Rubber; VIS, Viscose.

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